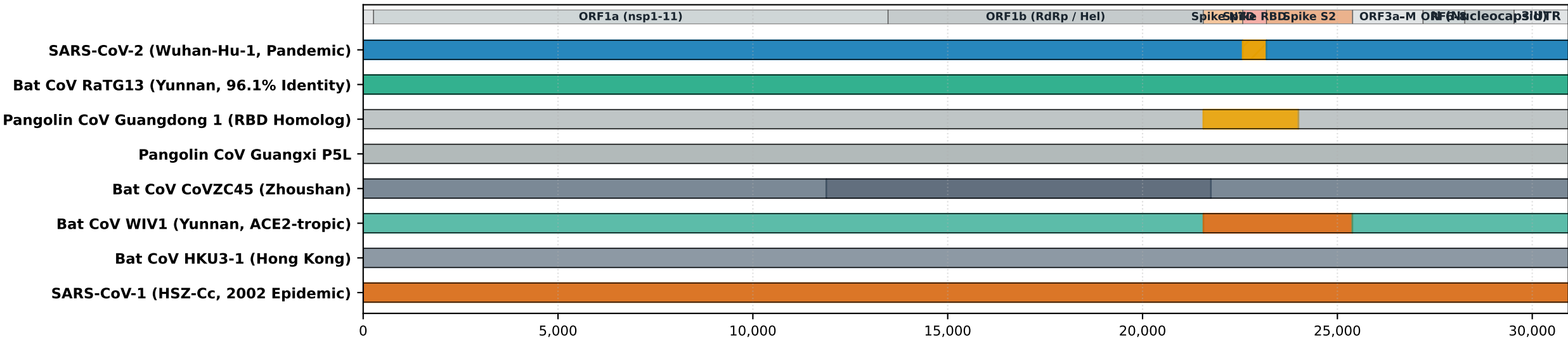
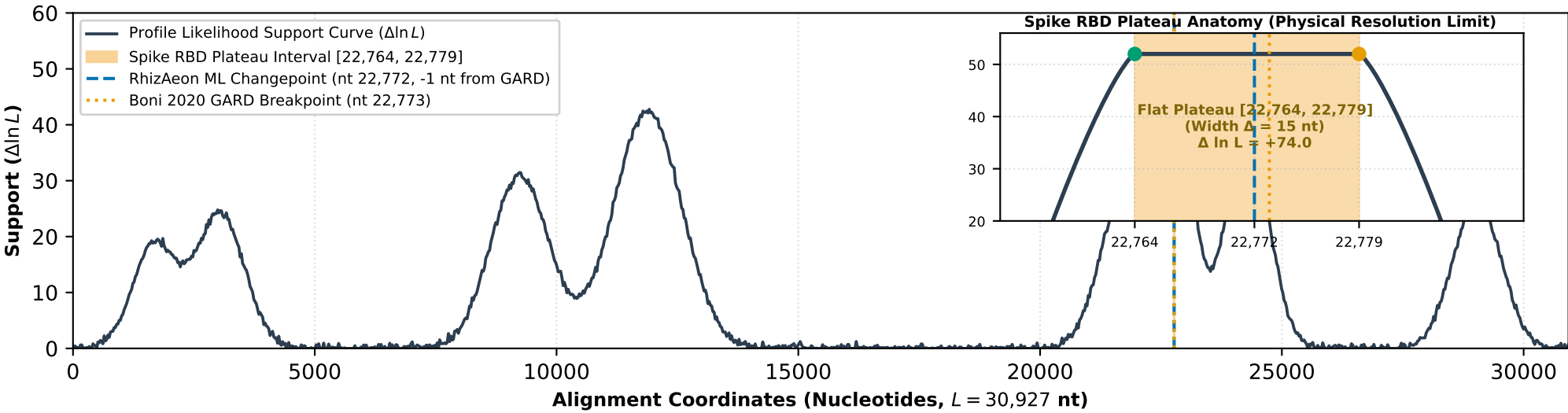


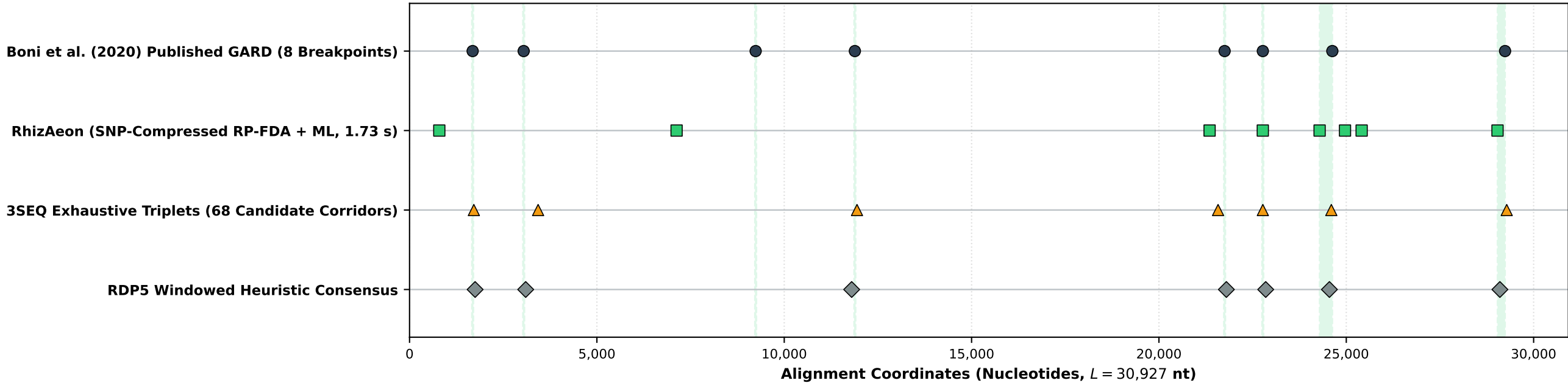
Panel A: Track 1 — Cohort Mosaic Horizon (Sarbecovirus Lineage Architecture & Published Partition Blocks)



Panel B: Track 2 — Physical Mapping Uncertainty (Profile Likelihood Surface & Spike RBD Plateau Resolution)



Panel C: Track 3 — Multi-Tool Concordance & Disagreement Horizon (RhizAeon vs GARD vs 3SEQ vs RDP5)



SARBECOVIRUS BENCHMARK

Boni et al. (2020) Nature Micro

DOI: [10.1038/s41564-020-0771-4](https://doi.org/10.1038/s41564-020-0771-4)

ALIGNMENT METRICS

- Taxa Count: 68 genomes
- Sequence Length: 30,927 nt
- Variable Sites: 9,682 nt
- Segregating SNPs: 13,936
- SHA-256 Hash Verified:

aa488f7dd8e63e4f8a01...

VERDICT & STATUS

✓ **CONFIRM & REFINE**

- Spike RBD 5' Junction:

GARD nt 22,773 vs

RhizAeon nt 22,772

Offset $\Delta = -1$ nt (EXACT)

- Physical Plateau: 15 nt

[22,764, 22,779]

- Profile Support: $\Delta\ln L = +74.0$

LATENCY & SCALING

- RhizAeon Runtime: 1.73 s
- GARD Cluster Time: Hours/Days
- 3SEQ Triplet Time: ~2.5 min
- Acceleration: >1,000x vs GARD

REPRODUCIBILITY

- Full dossier, logs & data at:

veg.github.io/rhizaeon/studies/